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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech (CSE) (CBCS) Degree Examinations
Supplementary Examination - December 2025
23CSE101 – ADVANCED DATA STRUCTURES & ALGORITHMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing One **full** question from each **Unit**.

		Unit - I	Marks	BT*	CO*	PO*
1.	a)	Given two sparse matrices: Matrix A: $\begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 7 \\ 0 & 6 & 0 & 0 \end{bmatrix}$ Matrix B: $\begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 4 \end{bmatrix}$ Represent both matrices using arrays for their sparse format and perform their addition.	8	L*3	1	2, 3
	b)	What do you mean by collision in hashing and explain how you can handle it by using linear probing with an example.	7	L2	1	3
	c)	Differentiate between normal array-based queue and linked list queue algorithm.	5	L3	1	2
2.	a)	Assume we have a one-dimensional array $a[\text{MAX_SIZE}]$, where the subscripts typically range from 0 to $\text{MAX_SIZE} - 1$. Using pointer arithmetic, how can we adjust the subscripts so that they range between -10 and 10, effectively creating an array that appears to have indices from -10 to 10? Explain the method with code for setting up such an array and accessing its elements using these subscripts.	8	L3	1	3
	b)	Given input {960, 971, 2110, 1402, 2342, 199, 2345, 1689} and a hash function $h(X) = X(\text{mod } 10)$, show the resulting: i. Separate chaining hash table ii. Open addressing hash table using linear probing iii. Open addressing hash table with quadratic probing iv. Open addressing hash table with second hash function $h_2(X) = 7 - (X \text{ mod } 7)$	8	L3	1	3
	c)	Write the four differences between doubly linked lists and circular linked lists.	4	L2	1	2, 3
		Unit - II				
3.	a)	What is a splay Tree. Explain how to insert an element from a Splay Tree considering all the cases and discuss the time complexity of this algorithm.	8	L3	2	3
	b)	What is a binary search tree (BST) and specify the steps showing the construction of a BST for the following data 10, 08, 15, 12, 13, 07, 09, 17, 20, 18, 04, 05.	8	L3	2	2, 3
	c)	What is an AVL Tree. Construct an AVL tree for the given list 12, 56, 89, 16, 25, 100, 75, 55, 95, 50.	4	L3	2	3



- 4 a) Traverse the given tree as shown in Figure Q4(a) with stepwise using Inorder, Preorder and Postorder traversals.

Given tree:

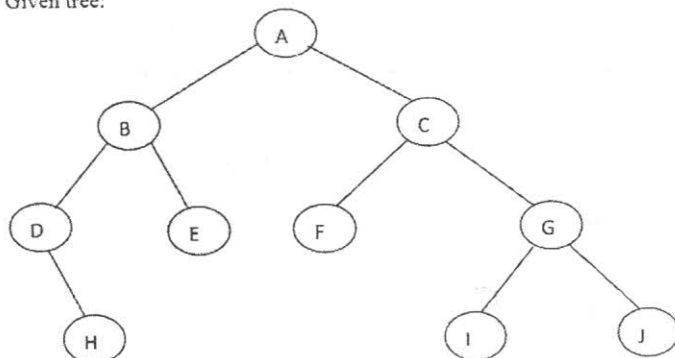


Figure: Q4(a)

- b) Explain clearly how to delete an element from a multiway Search Tree considering all the cases and discuss the time complexity of this algorithm.
 c) Explain about Left-Left Rotation & Right-Right Rotation in AVL tree with an example in stepwise.

9 L3 2 2,3

7 L3 2 3

4 L2 2 2

Unit - III

- 5 a) Analyze how the scalability of a polynomial time algorithm (e.g., $O(n^2)$) differs from that of an exponential time algorithm (e.g., $O(2^n)$).
 b) State and explain master Theorem with examples show how the Master Theorem is applied to solve recurrence relations for all 3 cases.
 c) What do you mean by asymptotic notations? Why is it used? Explain any two asymptotic notations.
- 6 a) Given the recurrence relation $T(n)=3T(n/4)+n^2$, apply the Recurrence Tree Method to construct the tree up to the third level. Calculate the total cost at each level and determine the overall time complexity.
 b) Illustrate the step-by-step process of the Substitution Method for solving recurrence relations.
 c) Analyze the four significances of the logarithmic function $\log n$ in the context of algorithm analysis.

8 L4 3 2,3

6 L2 3 2

6 L2 3 2

10 L3 3 2,3

6 L2 3 2

4 L2 3 2

Unit - IV

- 7 a) Consider the following graph given in Figure Q7(a) and apply Bellman-Ford algorithm to find the shortest path to reach all nodes in the given graph.

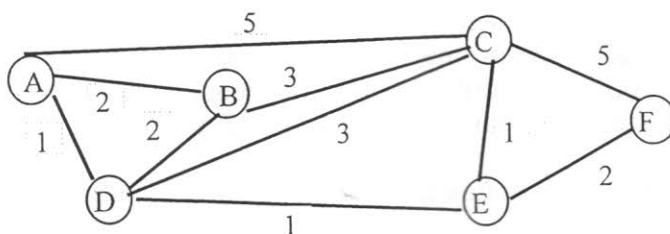


Figure : Q7(a)

- b) Explain Johnson's algorithm for finding the all-pairs shortest path.
 c) What is the significance of topological sorting in the process of finding the Single Source Shortest Path (SSSP) in a Directed Acyclic Graph (DAG)?

10 L3 4 3

5 L2 4 2

5 L3 4 3

8. a) Consider the following graph given in Figure Q8(a) and apply Dijkstra's algorithm to find the shortest path from source node **A** to all nodes in the given graph.

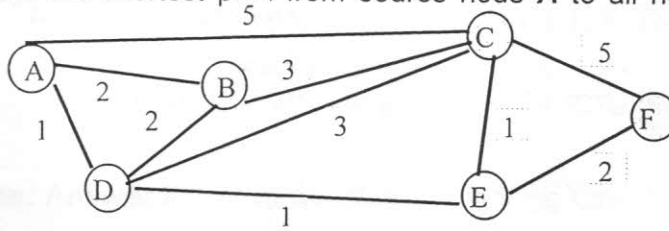
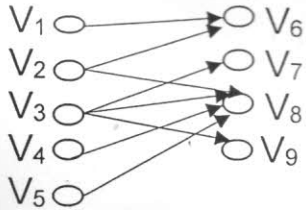


Figure: Q8(a)

- b) Solve maximum bipartite matching for the following graph 8 (b)



Graph 8(b)

- c) Compare the Ford Fulkerson algorithm with Johnson's algorithm.

10 L3 4 2,3

6	L3	4	3
4	L2	4	2

Unit - V

9. a) Explain the Naïve String-Matching algorithm and demonstrate how to apply it to locate the pattern 23415 within the text 235902341526739921.
 b) Describe how a transition function is constructed in a finite automaton for string matching with example.
10. a) Explain how this randomization could potentially reduce the average search time with example.
 b) Describe the Knuth-Morris-Pratt (KMP) string matching algorithm and demonstrate how to apply this algorithm to find the pattern xyzxyz within the text xyzxyzxxxxyzxyzxxxxyzzxyzxyz.

10 L3 5 3

10 L2 5 2

10 L2 5 2

10 L3 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
